Are We Really Measuring Forgetting? Pitfalls in Evaluating Machine Unlearning for Multimodal LLMs

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Abstract

As machine unlearning for multimodal large language models (MLLMs) rapidly advances, rigorous evaluation of its effectiveness becomes increasingly important. However, existing benchmarks largely inherit evaluation protocols from text-only unlearning and overlook the unique challenges of unlearning in multimodal settings. In this work, we identify three major pitfalls that may lead to misleading conclusions: the lack of heldout evaluation to distinguish genuine concept-wise forgetting from instance-wise overfitting, the reliance on synthetic, fine-tuned knowledge rather than real-world knowledge acquired during pre-training, and unfair method comparisons caused by applying the same training settings to different unlearning methods. To address these challenges, we introduce **GEMMU** (**G**eneralizable **E**valuation for **M**ultimodal **M**achine **U**nlearning), a benchmark built on real-world knowledge and a principled evaluation protocol. Re-evaluating representative unlearning methods under **GEMMU** reveals substantial generalization gaps and challenges conclusions drawn from previous benchmarks. Our findings highlight the importance of rigorous evaluation for advancing multimodal machine unlearning and provide guidelines for future benchmark design.

1 Introduction

As MLLMs (Bai et al., 2025; Li et al., 2025) are increasingly deployed in real-world applications, concerns surrounding privacy and data governance have intensified. In this context, machine unlearning has attracted growing attention as a mechanism for removing sensitive or private knowledge, such as personal information, from trained models. As unlearning methods advance, rigorous evaluation is essential to reliably assess their effectiveness.

Several benchmarks (Liu et al., 2025a; Dontsov et al., 2025; Wang et al., 2025) have been proposed to assess progress in multimodal machine unlearning. However, we identify the following fundamental pitfalls in existing benchmarks that may lead to overly optimistic or misleading evaluations.

First, testing on the training set is widely recognized as poor practice in machine learning, yet existing unlearning protocols often evaluate forgetting and retention on the same data used during unlearning (Liu et al., 2025a; Ma et al., 2025; Zheng et al., 2026). Such evaluation risks overestimating unlearning effectiveness, as performance on these examples may not generalize to unseen data. This concern is particularly important in the multimodal setting, where effective unlearning should remove knowledge at the *concept level* rather than merely suppressing specific visual *instances*.

Second, many existing benchmarks (Dontsov et al., 2025; Wang et al., 2025; Liu et al., 2025a; Xu et al., 2025) focus primarily on newly fine-tuned knowledge, rather than real-world knowledge acquired during pre-training. While convenient for evaluation, such settings may not reflect realistic unlearning scenarios. Moreover, existing unlearning methods largely rely on reverse fine-tuning, which may be more effective at removing newly introduced knowledge. In contrast, pre-trained knowledge is often deeply entangled with other useful knowledge, making it difficult to remove selectively without affecting the model's general capabilities. As a result, those benchmarks may overestimate the effectiveness of existing methods in real-world unlearning.

Third, most existing benchmarks evaluate different unlearning methods using identical hyperparameters, despite differences in their optimization objectives and training dynamics (Xu et al., 2025; Dontsov et al., 2025). A configuration that works well for one method may be suboptimal for another. Consequently, conclusions about which method is more effective can depend on the shared configuration rather than on the methods themselves.

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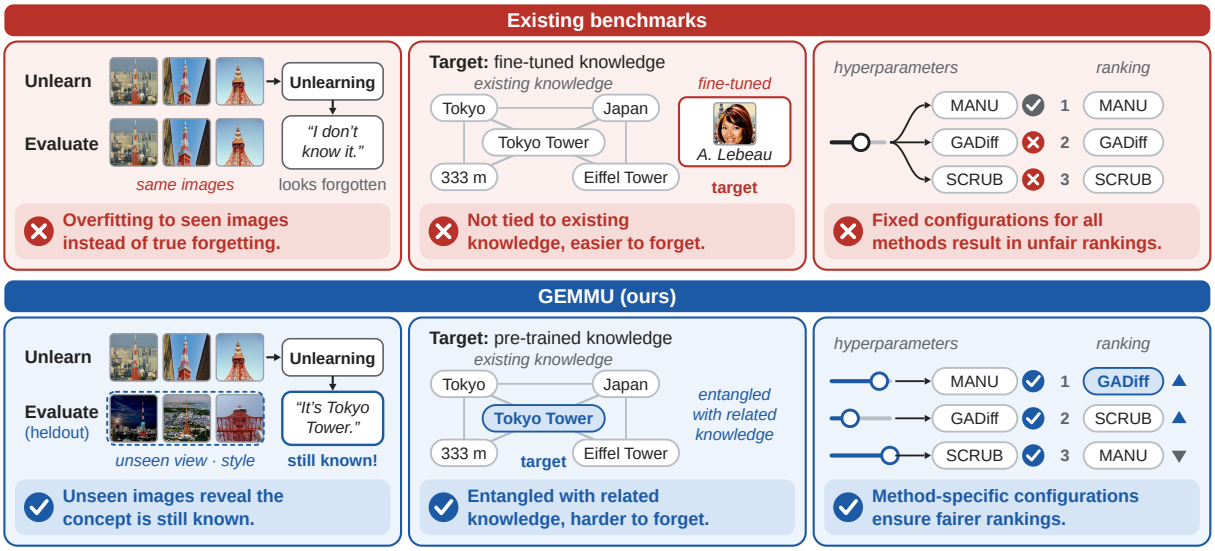

Figure 1: Three major pitfalls in existing multimodal unlearning benchmarks (top) and how GEMMU addresses them (bottom) through heldout evaluation, real-world knowledge unlearning, and method-specific hyperparameter tuning to achieve generalizable evaluations.

To address these limitations, we introduce **GEMMU** (Generalizable Evaluation for Multimodal Machine Unlearning; Fig. 1), a benchmark built on three principles: heldout evaluation for both forget and retain sets, unlearning of real-world pre-trained knowledge, per-method hyperparameter search (Tab. 1), and additionally modality gap analysis (Sec. 5.4). Re-evaluating representative unlearning methods with **GEMMU** reveals substantial generalization gaps overlooked by existing benchmarks, shows that method rankings can change under different configurations, and calls into question the effectiveness of current methods. Our main contributions are summarized as follows:

- We identify major pitfalls in existing MLLM unlearning evaluations: *testing on the training set*, *reliance on fine-tuned knowledge*, and *unfair hyperparameter settings* across methods.
- We introduce **GEMMU**, a more rigorous benchmark that addresses these limitations and provides principles for future benchmark design.
- We re-evaluate nine representative unlearning methods and uncover substantial generalization gaps and ranking changes overlooked by previous benchmarks.

2 Related Works

Machine Unlearning Methods for MLLMs Existing approaches to multimodal machine unlearning mainly follow a *reverse fine-tuning* paradigm, typically suppressing forget-set knowledge through gradient ascent while preserving retained knowledge through gradient descent or other regularization objectives. Representative methods include GA (Thudi et al., 2022), GADiff (Liu et al., 2022), KL Min and IDK (Maini et al., 2024), NPO (Zhang et al., 2024), LLMU (Yao and Xu, 2024), RMU (Li et al., 2024b), and SCRUB (Kurmanji et al., 2023). While some of the above methods were originally developed for text-only LLMs, they have served as common baselines for unlearning in MLLMs. More recently, model editing has emerged as an alternative to conventional tuning-based unlearning. For example, MANU (Liu et al., 2025b) selectively identifies and prunes model components associated with the knowledge to be forgotten.

Machine Unlearning Evaluation for MLLMs Early language-only benchmarks such as TOFU (Maini et al., 2024) establish a controlled evaluation setting for fictitious knowledge removal, and MLLMU-Bench (Liu et al., 2025a) extends this direction to multimodal models. CLEAR (Dontsov et al., 2025), FIUBench (Ma et al., 2025), UMU-Bench (Wang et al., 2025), and PEBench (Xu et al., 2025) introduce more comprehensive multimodal evaluation settings covering person profiles, cross-modal knowledge, and personal entities with event scenes. Despite this progress, existing benchmarks remain limited for evaluating practical multimodal unlearning.

Benchmark	Proper Train-test Split	Knowledge Acquisition	Unlearned Target Source	# Real-world Concepts	Concept Types	Hyperparameter Selection	Open-source
MMUBench (Li et al., 2024a)	Partial	Pre-trained	Real-world	20	Diverse	One-size-fits-all	No
MLLMU-Bench (Liu et al., 2025a)	No	Fine-tuned	Synthetic	–	Human only	One-size-fits-all	Yes
FIUBench (Ma et al., 2025)	No	Fine-tuned	Synthetic	–	Human only	Per-method	Yes
CLEAR (Dontsov et al., 2025)	No	Fine-tuned	Synthetic	–	Human only	One-size-fits-all	Yes
PEBench (Xu et al., 2025)	Partial	Fine-tuned	Synthetic	–	Humans, events	One-size-fits-all	Yes
UMU-Bench (Wang et al., 2025)	No	Fine-tuned	Synthetic	–	Human only	One-size-fits-all	Yes
PULSE (Kawakami et al., 2025)	No	Pre-trained	Real-world	20	Human only	Not listed	No
OFFSIDE (Zheng et al., 2026)	No	Fine-tuned	Real-world	80	Human only	Per-method	Yes
GEMMU (Ours)	Yes	Pre-trained	Real-world	**139**	Diverse	Searched	Yes

Table 1: Comparison of existing multimodal machine unlearning benchmarks. **GEMMU** is the most comprehensive benchmark that evaluates the unlearning of real-world, pre-trained knowledge with a proper train-test split.

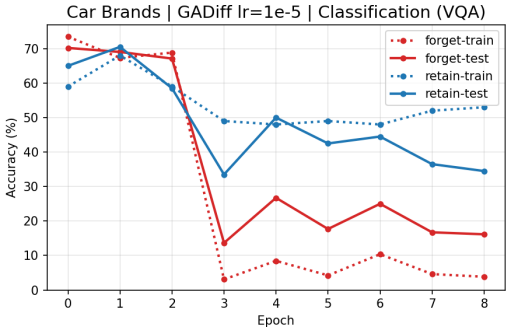

Figure 2: A representative example of the generalization gap. The red dotted curve falls well below the red solid curve, while the blue dotted curve remains above the blue solid curve, revealing substantial train–test discrepancies after unlearning.

They still rely on generated identities, which require fine-tuning to memorize this new knowledge and may not reflect the unlearning of real-world, pre-trained knowledge. MMUBench and PEBench (Li et al., 2024a; Xu et al., 2025) partition the forget set into training and test examples but overlook the same problem for the retain set. An existing preprint, PULSE (Kawakami et al., 2025), studies the unlearning of pre-trained knowledge, but does not provide a theoretical analysis of the distinction between fine-tuned and pre-trained knowledge, nor does it address overfitting to the forget or retain sets.

3 Preliminaries

Formulation of the Unlearning Task Let f_θ denote an MLLM with parameters θ. Following the standard formulation in this field, the unlearning task considers two datasets: a forget set $\mathcal{D}_{\text{forget}}$, containing knowledge to be removed, and a retain set $\mathcal{D}_{\text{retain}}$, containing knowledge to be preserved. Each dataset consists of image-question-answer triplets (I, q, a), where I is empty for text-only questions. We further partition both datasets into disjoint training and test sets to evaluate generalizability: $\mathcal{D}^{\text{train}}_{\text{forget}}$, $\mathcal{D}^{\text{test}}_{\text{forget}}$, $\mathcal{D}^{\text{train}}_{\text{retain}}$ and $\mathcal{D}^{\text{test}}_{\text{retain}}$.

Given a pre-trained MLLM f_{θ_0}, where θ_0 denotes the parameters before unlearning, an unlearning method $\mathcal{U}$ updates θ_0 using the training splits:

$$\theta_u = \mathcal{U}(\theta_0, \mathcal{D}^{\text{train}}_{\text{forget}}, \mathcal{D}^{\text{train}}_{\text{retain}}),$$

where θ_u denotes the parameters after unlearning. Note that previous works often use the entire $\mathcal{D}_{\text{forget}}$ and $\mathcal{D}_{\text{retain}}$ in place of the training splits mentioned above. The objective is to reduce the model's ability to recover knowledge in $\mathcal{D}_{\text{forget}}$, while preserving its performance on $\mathcal{D}_{\text{retain}}$. Successful unlearning should achieve these goals not only on the training splits but also on the test splits, which provide a more reliable measure of generalizable forgetting and knowledge retention.

4 Pitfalls in Past Evaluations

We discuss three major pitfalls in existing multimodal unlearning benchmarks: the lack of proper train-test splits for concept-wise unlearning, the reliance on synthetic, fine-tuned knowledge, and unfair comparisons arising from applying identical hyperparameters across different unlearning methods. All experiments are conducted on our proposed **GEMMU** dataset, with construction details provided in Sec. 5.

4.1 Testing on the Training Set Is Problematic for Concept-wise Unlearning

A common practice in prior multimodal unlearning evaluations (Liu et al., 2025a; Huo et al., 2025; Dontsov et al., 2025) is to evaluate forgetting on the same $\mathcal{D}_{\text{forget}}$ used during unlearning, and similarly evaluate retention on the same $\mathcal{D}_{\text{retain}}$. Unfortunately, this violates one of the most fundamental

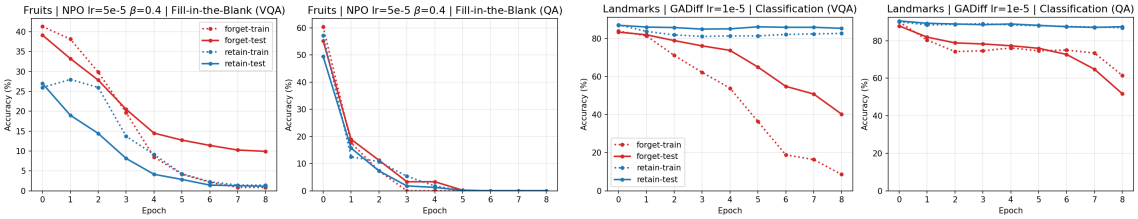

Figure 3: Comparison of unlearning generalization between image-text (left) and text-only (right) inputs. Image-text unlearning exhibits a substantially larger generalization gap than text-only unlearning, despite using the same dataset and training configurations.

principles of machine learning: evaluating generalization on unseen data. A model may appear to forget the target knowledge and retain the rest, while merely overfitting to the examples seen during training. Fig. 2 shows a representative example of overly optimistic training-set evaluation, where accuracy is lower on the forget set and higher on the retain set compared to their respective test sets.

One might argue that some prior works additionally evaluate general capabilities on heldout, real-world benchmarks. However, such evaluations do not address the fundamental issue. Without proper train-test splits for both $\mathcal{D}_{\text{forget}}$ and $\mathcal{D}_{\text{retain}}$, the reported forgetting and retention scores remain unreliable indicators of unlearning effectiveness, despite being the primary metrics.

Why is this issue more important in multimodal unlearning? In some text-only unlearning scenarios, such as forgetting specific credit card numbers or passwords, the target is an exact string, making training-set evaluation potentially adequate for such **instance-wise unlearning**. In contrast, the same visual concept (e.g., a photo of Tokyo Tower) can appear in images with different viewpoints, backgrounds, and styles. This introduces greater variation in the input space, making generalization particularly important for **concept-wise unlearning**. Fig. 3 shows examples where image-text unlearning (concepts provided in images) exhibits a substantially larger generalization gap than text-only unlearning (concepts provided in pure text), highlighting the risk of overestimating forgetting in multimodal machine unlearning when evaluating only on training examples.

4.2 Reliance on Synthetic, Fine-tuned Knowledge

Another key limitation of existing multimodal unlearning benchmarks is their reliance on knowledge introduced through fine-tuning, often in the form of synthetic facts or identities. While such settings offer a controlled environment for evaluating unlearning, they may not accurately reflect the challenges of real-world knowledge removal. In particular, unlearning fine-tuned knowledge and pre-trained knowledge may differ substantially in their underlying dynamics and difficulty.

Synthetic knowledge introduced through fine-tuning often consists of relatively isolated facts that are less entangled with the model's existing knowledge. *Reverse fine-tuning*, which underlies many existing unlearning algorithms, may provide a relatively straightforward way to undo the effects of fine-tuning and thereby achieve unlearning. In contrast, real-world, pre-trained knowledge is acquired through extensive pre-training and may be deeply entangled with other useful knowledge, making its removal more challenging. As a result, strong performance on synthetic or fine-tuned knowledge does not necessarily translate to effective unlearning of real-world, pre-trained knowledge.

To better understand the differences between unlearning pre-trained and fine-tuned knowledge, we investigate both settings from two perspectives: (1) learning dynamics and (2) the forgetting–retention trade-off. These two perspectives provide complementary empirical and theoretical insights into unlearning difficulty.

Learning Dynamics. We compare the two settings directly by tracking forget-set performance over unlearning epochs on two matched pairs of GEMMU subsets: real faces versus synthetic identities, and real landmarks versus fictitious ones, where the second subset of each pair is injected into the model through fine-tuning.

In Fig. 4, fine-tuned knowledge drops sharply within the first epoch, whereas pre-trained knowledge decays much more slowly on both LLaVA-1.5-7B and Qwen3-VL-8B-Instruct. Plots for other unlearning algorithms are provided in Appendix A.

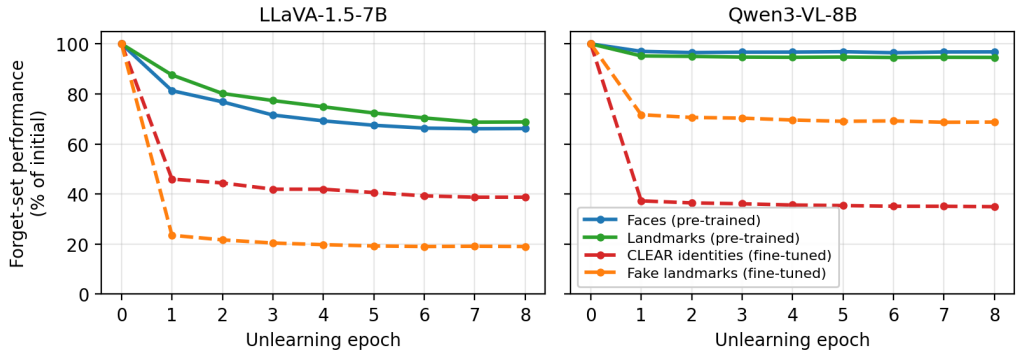

Figure 4: Forget-set image-textual performance (relative to the model before unlearning) over NPO unlearning epochs under identical hyperparameters, on LLaVA-1.5-7B (left) and Qwen3-VL-8B-Instruct (right). Fine-tuned knowledge (dashed) collapses much more quickly than pre-trained knowledge (solid).

Forgetting–Retention Trade-off. Another way to compare unlearning fine-tuned and pre-trained knowledge is to examine the retention cost required to achieve a given level of forgetting. To formalize this intuition, we present Proposition 1.

Definitions. Let $s_{\text{target}}(\theta)$ and $s_{\text{heldout}}(\theta)$ denote knowledge scores for the target and heldout concepts, respectively, with higher values indicating better performance. We say that forgetting is successful when s_{target} is below a certain threshold τ, and define the required amount of forgetting as $\gamma = s_{\text{target}}(\theta_0) - \tau > 0$. The retention cost is then $s_{\text{heldout}}(\theta_0) - s_{\text{heldout}}(\theta_u)$.

Proposition 1 (Forgetting–Retention Trade-off)

I. For fine-tuned knowledge, zero retention cost is attainable by reverting to the original model parameters θ_0.

II. For pre-trained knowledge, an unlearned model θ_u satisfying the forgetting criterion and assumptions incurs a strictly positive retention cost:

$$s_{\text{heldout}}(\theta_0) - s_{\text{heldout}}(\theta_u) \geq \frac{\mu\gamma^2}{2K^2} - \frac{\varepsilon_0\gamma}{K} > 0.$$

under the assumptions specified in Appendix B.2. Here, K, μ, and ε_0 are constants defined in Appendices B.1 and B.2.

The proof is provided in Appendix B. This result suggests that unlearning pre-trained knowledge can be viewed as more challenging than unlearning fine-tuned knowledge, as successful forgetting necessarily comes at the cost of other useful knowledge under our assumptions.

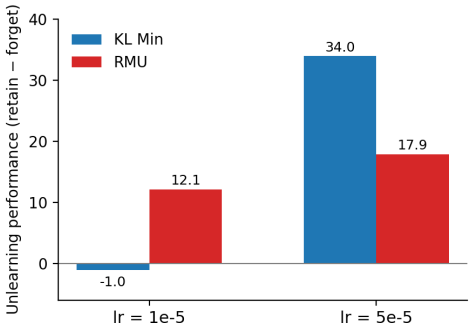

Figure 5: KL Min and RMU under two shared learning rates on LLaVA-1.5-7B. The winner flips with the shared configuration, showing that one-size-fits-all comparisons are unreliable.

4.3 Hyperparameter Sensitivity

Most existing benchmarks compare unlearning methods under a single shared training configuration (Tab. 1). This implicitly assumes that different methods share a common operating range, which we find to be far from true. [1]

Fig. 5 compares KL Min and RMU trained under identical shared configurations on LLaVA-1.5-7B, measured by average retain-set minus average forget-set performance on heldout examples, where higher means better forgetting at lower retention cost. At a learning rate of 10^{-5}, RMU wins clearly while KL Min appears entirely ineffective; at 5×10^{-5}, the conclusion reverses, with KL Min now outperforming RMU by a wide margin. The same shared protocol thus supports two contradictory conclusions about which algorithm is better.

Fig. 6 extends this observation to all nine meth-

[1] For benchmarks that do not report hyperparameter selection in the paper, we classify them by the hyperparameters in their released code.

Methods	Forget Test (↓)						Retain Test (↑)						Rank
	Classification		FITB		Generation		Classification		FITB		Generation		
	VQA	QA	VQA	QA	VQA	QA	VQA	QA	VQA	QA	VQA	QA	
LLaVA-1.5-7B													
Pre-trained	74.86	79.89	51.59	70.92	15.77	70.47	72.29	80.00	40.38	65.79	13.26	68.81	–
GA (Thudi et al., 2022)	64.76	65.87	20.79	39.06	7.70	19.28	60.97	66.37	18.97	44.93	6.95	19.45	8
GADiff (Liu et al., 2022)	**4.33**	**0.00**	2.55	**0.00**	3.20	**0.00**	48.49	69.29	54.61	87.52	**34.18**	**81.50**	1
SCRUB (Kurmanji et al., 2023)	19.17	75.19	9.27	51.90	3.87	60.48	69.50	**80.03**	59.04	**87.81**	29.52	80.75	4
KL Min (Maini et al., 2024)	47.26	30.81	23.81	14.52	7.13	18.24	68.87	79.31	41.87	70.96	14.28	70.42	3
NPO (Zhang et al., 2024)	72.08	79.34	33.57	43.40	13.31	22.83	69.18	77.93	30.12	51.05	12.49	23.45	9
IDK (Maini et al., 2024)	60.81	71.82	6.10	34.81	3.30	42.52	**69.69**	76.26	**87.80**	85.45	33.85	72.82	2
LLMU (Yao and Xu, 2024)	66.84	73.50	23.85	24.02	5.96	39.10	65.06	77.05	21.59	29.97	6.94	42.53	7
RMU (Li et al., 2024b)	4.54	74.11	**1.16**	64.13	**1.06**	54.63	65.74	75.86	34.34	58.02	18.21	54.88	5
MANU (Liu et al., 2025b)	60.14	70.35	20.68	37.52	5.11	58.41	63.78	74.75	25.99	43.55	7.29	61.81	6
Qwen3-VL-8B-Instruct													
Pre-trained	88.82	86.32	82.61	78.37	42.11	59.49	86.69	85.25	76.39	78.20	43.16	59.23	–
GA (Thudi et al., 2022)	67.76	65.97	34.62	38.45	3.76	8.53	64.83	67.90	28.28	37.46	4.76	12.41	8
GADiff (Liu et al., 2022)	13.91	**31.10**	**0.10**	**0.00**	**0.23**	**0.00**	56.70	68.09	58.62	57.46	**53.01**	73.35	2
SCRUB (Kurmanji et al., 2023)	3.07	41.22	1.39	31.34	0.80	20.56	84.11	87.04	76.03	**86.24**	44.72	**73.97**	1
KL Min (Maini et al., 2024)	88.81	86.10	79.60	73.65	40.11	57.38	**86.76**	85.12	77.41	78.31	42.20	59.41	7
NPO (Zhang et al., 2024)	68.63	66.17	22.06	43.92	3.10	10.58	65.44	68.37	19.48	40.31	2.43	11.26	9
IDK (Maini et al., 2024)	65.17	86.38	7.83	38.88	2.66	13.73	53.84	**90.43**	**86.92**	67.09	42.53	62.25	3
LLMU (Yao and Xu, 2024)	86.17	83.76	34.80	42.65	19.70	25.73	84.78	83.94	35.32	49.97	18.56	27.38	6
RMU (Li et al., 2024b)	**2.48**	83.91	1.66	76.88	4.02	59.22	84.34	85.28	71.44	75.18	41.01	58.98	4
MANU (Liu et al., 2025b)	49.65	35.62	21.37	20.26	34.68	41.02	76.31	65.08	48.80	46.02	40.64	55.26	5

Table 2: Benchmark results on LLaVA-1.5-7B and Qwen3-VL-8B-Instruct, averaged over the four real-world subsets. VQA denotes image-textual and QA pure-text evaluation. **Bold** and underline denote the best and second-best result per column within each model; Rank orders methods by average retain-test minus average forget-test performance. The classic and simple GADiff achieves the strongest overall balance between forgetting and retention.

ods across our search space. Without proper tuning, many configurations lie on the $y = x$ line, i.e., indiscriminate forgetting. After per-method search, the strongest methods fall well below it. This sensitivity partly reflects that multimodal unlearning is harder than text-only unlearning, where a shared configuration is common practice (Maini et al., 2024), and further indicates that fair evaluation for multimodal unlearning requires a per-method hyperparameter search. More examples are provided in Appendix D.

5 GEMMU

GEMMU is designed to reveal the pitfalls discussed in Sec. 4 and to provide practical guidelines that future evaluation benchmarks should follow. Its core components include (1) concept removal of existing real-world knowledge and comparison with fine-tuned knowledge; (2) generalization of forgetting and retention to heldout examples; (3) per-method hyperparameter search for fair comparisons; and additionally, (4) modality gap analysis using image-text and text-only questions, inspired by Wang et al. (2025) and Huo et al. (2025). We first describe our methodology (Sec. 5.1), followed by a re-benchmarking of existing unlearning methods (Sec. 5.4), and conclude with design principles for future benchmarks (Sec. 5.5).

5.1 Benchmark Construction

Concept removal for real-world and synthetic knowledge. To achieve a diverse source of real-world knowledge, we consider human identities, landmarks, car brands, and fruits as our main categories, with 139 concepts in total. Following Liu et al. (2025a) and Wang et al. (2025), each concept is assigned exclusively to the forget or retain set. We also verified that the evaluated MLLMs have a decent understanding of those concepts before unlearning in Tab. 2.

To compare the unlearning of pre-trained and fine-tuned knowledge in Sec. 4.2, we additionally include two synthetic knowledge subsets: CLEAR and fake landmarks. For CLEAR, we adopt fictitious identities, along with their associated images and biographical information from the CLEAR dataset (Dontsov et al., 2025). For fake landmarks, we generate fictional landmark profiles with cor-

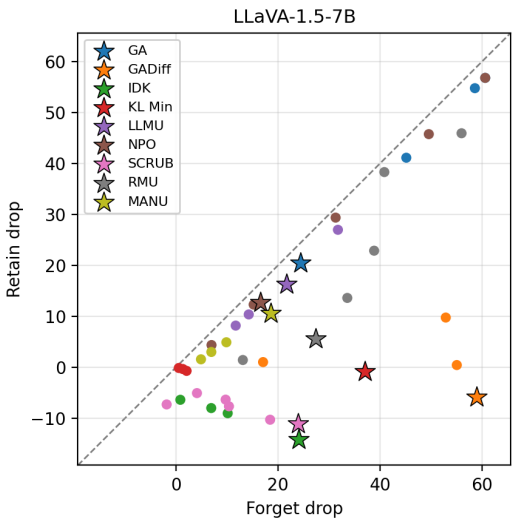

Figure 6: Forget-test drop versus retain-test drop for every configuration in our search space on LLaVA-1.5-7B. Without proper tuning, many configurations lie on the $y = x$ line, where retained knowledge is damaged as much as the target is forgotten.

Methods	LLaVA-1.5-7B		Qwen3-VL-8B		Removal type
	VQA drop (%)	QA drop (%)	VQA drop (%)	QA drop (%)	
GA	34.4	43.9	50.3	49.6	Whole concept
GADiff	92.9	100.0	93.3	86.1	Whole concept
SCRUB	77.3	15.2	97.5	58.5	Association
KL Min	45.0	71.3	2.4	3.1	Inconsistent
NPO	16.4	34.2	56.1	46.2	Whole concept
IDK	50.6	32.6	64.6	38.0	Association
LLMU	32.0	38.3	34.1	32.1	Whole concept
RMU	95.3	12.8	96.2	1.9	Association
MANU	39.6	24.9	50.5	56.8	Whole concept

Table 3: Forget-test performance drop relative to the pre-trained model (% of initial), per modality, averaged over the four real-world subsets at each method's selected configuration. Association removers suppress VQA while QA largely survives while whole-concept removers suppress both.

responding images and construct questions about their attributes, including location and completion date. Each synthetic subset consists of 40 concepts.

Generalizability analysis. Within each concept, examples are split into disjoint training and test sets, so the test sets assess unseen examples of the same concepts. Images are collected from the Internet and manually verified by the authors.[2] We adopt a train-test split of 5:10, which is less extreme than the single-training-data setup in MMUBench (Li et al., 2024a).

[2]We will release the source links instead of the images to prevent copyright infringement.

Per-method hyperparameter search. For each method, we search for its optimal hyperparameters by scoring each configuration by its average retain-train minus average forget-train performance and select the configuration with the best score averaged over the four real-world subsets. Selection uses training splits only so that test data never influences it. See Appendix C for the selected hyperparameters and implementation details.

Modality gap analysis. Following Liu et al. (2025a) and Dontsov et al. (2025), we construct questions in two input modalities: visual question answering (VQA) pairs an image with a textual question, while text-only question answering (QA) contains the concept name and no image. Both modalities cover single-choice classification, fill-in-the-blank, and generation tasks. We use accuracy for classification and fill-in-the-blank tasks and ROUGE-L for generation. For both image-text and text-only tasks, we paraphrase the question texts while preserving their meaning and ground-truth answers to prevent overfitting to the question formats.

Representative examples from the forget and retain training and test sets of all six subsets are provided in Appendix E.1.

5.2 Participating MLLMs

We consider Qwen3-VL-8B-Instruct (Bai et al., 2025) and LLaVA-1.5-7B (Liu et al., 2024) as our base MLLMs. Following prior works on multimodal unlearning (Huo et al., 2025; Dontsov et al., 2025; Zheng et al., 2026), we adopt LoRA (Hu et al., 2022) for both fine-tuning the base models in the synthetic versus pre-trained knowledge experiment and applying the unlearning algorithms.

5.3 Machine Unlearning Methods

We evaluate nine representative machine unlearning methods covering a diverse range of unlearning strategies. Four are built on *gradient ascent* over forget-set responses: GA (Thudi et al., 2022) uses the ascent loss alone; GADiff (Liu et al., 2022) adds gradient descent on the retain set; KL Min (Maini et al., 2024) instead regularizes retain updates with a KL penalty toward the pre-trained model; and LLMU (Yao and Xu, 2024) further pairs forget prompts with unrelated responses under a KL penalty. NPO (Zhang et al., 2024) replaces unbounded ascent with a preference-style loss whose ascent weight shrinks as the forget loss exceeds

the pre-trained loss. IDK (Maini et al., 2024) simply relabels forget answers with abstention targets. SCRUB (Kurmanji et al., 2023) is *distillation-based*, maximizing KL divergence from a frozen teacher on forget examples while minimizing it on retain examples. Finally, two methods operate on *internal representations*: RMU (Li et al., 2024b) steers forget-set activations at a chosen layer toward a random vector while anchoring retain-set activations, and MANU (Liu et al., 2025b) prunes MLP neurons ranked by their forget-versus-retain importance.

5.4 Re-benchmarking Unlearning Algorithms

Table 2 shows results for nine unlearning methods under **GEMMU**. Our main findings are as follows.

Simple methods win. Recent methods are often reported as state-of-the-art on prior benchmarks. For example, MANU (Liu et al., 2025b) outperforms GADiff on MLLMU-Bench (Liu et al., 2025a). Rankings on **GEMMU** differ: the two strongest methods are among the oldest and simplest, GADiff and SCRUB.

Association removal vs. concept removal. The two modalities reveal *what* is unlearned. Following the notion in Huo et al. (2025), if only VQA drops while QA survives, the model has merely severed the association between visual appearance and knowledge; if both drop, the concept itself is removed. Tab. 3 classifies each method by comparing its VQA and QA drops on the forget-test split based on the results in Tab. 2. RMU is the clearest association remover: on Qwen3-VL-8B, VQA classification falls to 2.48 while QA stays at 83.91 (FITB: 1.66 vs. 76.88), so the model no longer recognizes the visual entity but still knows everything about it. GADiff removes whole concepts, at the cost of the largest retain-side drop.

No method solves the task. Even the best entries in Table 2 fall short of ideal unlearning: GADiff sacrifices retain classification (48.49 on LLaVA, 56.7 on Qwen), SCRUB leaves textual forget knowledge largely intact, and the remaining methods either under-forgets or over-destroys. We conclude that unlearning real-world, pre-trained multimodal knowledge remains unsolved.

5.5 On Guidelines for Future Multimodal Unlearning Benchmarks

We close by distilling our findings into design guidance, in the hope that good practices are inherited by future benchmarks.

What makes a good design? First, the fundamentals of machine learning evaluation: a proper train–test split, so that forgetting and retention are measured as generalization rather than memorization (Sec. 4.1), and a hyperparameter search per method, without which conclusions about relative effectiveness may reflect the chosen configuration rather than the methods themselves (Sec. 4.3). Second, echoing modality-gap studies in multimodal unlearning (Wang et al., 2025; Huo et al., 2025), benchmarks should construct and report both VQA and QA metrics for every task. Only the contrast between the two reveals whether a method removes the visual–textual association or the concept itself (Sec. 5.4).

What to avoid? Inheriting text-only practices without accounting for what is unique to MLLMs: instance-wise forgetting may be defensible for text, but not for concept-wise forgetting in MLLMs (Fig. 3). Likewise, convenience should not trump rigor. Benchmarks built on synthetic, fine-tuned knowledge are easier to construct but may not reflect realistic unlearning (Sec. 4.2), and applying one shared hyperparameter configuration to every method can distort the comparison (Sec. 4.3).

6 Conclusion

In this work, we introduce **GEMMU**, a benchmark for multimodal machine unlearning that emphasizes generalization-aware evaluation and covers a broader and more diverse range of real-world concepts than prior benchmarks. Experiments on **GEMMU** reveal substantial generalization gaps overlooked by previous benchmarks, highlight the challenges of removing pre-trained knowledge that is deeply entangled with retained knowledge, and demonstrate how proper hyperparameter tuning can alter the relative rankings of unlearning methods.

As unlearning methods and benchmarks become increasingly sophisticated, we emphasize that fundamental principles of evaluation must not be overlooked to ensure that scientific progress is built on a solid foundation.

Limitations

Following common practice in existing multimodal unlearning work, we use LoRA (Hu et al., 2022) for both fine-tuning and unlearning. Full-model fine-tuning may exhibit different unlearning behavior from adapter-based fine-tuning. However, we do not extend our experiments to full-model fine-tuning due to its substantially higher computational cost, and instead focus on settings that can be feasibly evaluated on an NVIDIA H100 GPU.

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A Extended Results for Learning Dynamics

Fig. 7 shows the learning dynamics averaged across all eight learning-based unlearning methods. The findings in Sec. 4.2 remain consistent, with pre-trained knowledge being forgotten more slowly than fine-tuned knowledge. The gap is smaller in the averaged results because some unlearning methods either fail to induce meaningful forgetting or cause the model to collapse entirely.

B Proof of Proposition 1

B.1 Analysis setup

We use differentiable knowledge scores s_{target} and s_{heldout} for the analysis, such as the average log-likelihood of reference answers. Let Θ be a neighborhood containing θ_0 where the assumptions below hold, and let $\|\cdot\|_2$ denote the Euclidean norm. Consider any updated model $\theta_u \in \Theta$ satisfying the forgetting criterion $s_{\text{target}}(\theta_u) \le \tau$.

We measure retention cost relative to the original model as $s_{\text{heldout}}(\theta_0) - s_{\text{heldout}}(\theta_u)$. For fine-tuned knowledge, we show that $\theta_u = \theta_0$ satisfies the forgetting criterion with zero retention cost. For pre-trained knowledge, we establish a strictly positive lower bound on the retention cost under the following assumptions.

B.2 Assumptions

The following conditions are needed for the pre-trained case. Here, the target concept is already present in θ_0, and

$$\gamma := s_{\text{target}}(\theta_0) - \tau > 0$$

is the score reduction required to reach the forgetting threshold.

Assumption A1 (Lipschitz target score) *s_{target} is K-Lipschitz, i.e., there is $K > 0$ such that, for all $\theta, \theta' \in \Theta$,*

$$|s_{\text{target}}(\theta) - s_{\text{target}}(\theta')| \le K\|\theta - \theta'\|_2.$$

This condition limits how much the target score can change per unit of parameter displacement. Reducing the score by at least γ therefore requires a displacement of at least γ/K.

Assumption A2 (Limited local improvement) *At θ_0, we bound how quickly retained utility can improve under small parameter changes: for some $\varepsilon_0 \ge 0$,*

$$\|\nabla s_{\text{heldout}}(\theta_0)\|_2 \le \varepsilon_0.$$

This reflects the assumption that the pre-trained model is near-optimal on the heldout concepts. Even in the most favorable direction, an update δ provides at most $\varepsilon_0\|\delta\|_2$ of improvement under the linear approximation.

Assumption A3 (Quadratic penalty) *There is $\mu > 0$ such that, for every $\theta_0 + \delta \in \Theta$,*

$$s_{\text{heldout}}(\theta_0 + \delta) \le s_{\text{heldout}}(\theta_0) + \langle \nabla s_{\text{heldout}}(\theta_0), \delta \rangle - \frac{\mu}{2}\|\delta\|_2^2.$$

This condition formalizes the intuition that pre-trained knowledge is entangled with other useful knowledge: it places retained utility below its first-order approximation by at least $\mu\|\delta\|_2^2/2$ for every update within Θ. The penalty grows quadratically with displacement, whereas the possible first-order improvement in Assumption A2 grows only linearly. Together, they imply a positive retention cost when the displacement is sufficiently large.

To ensure that successful forgetting requires such a displacement, we additionally require

$$\varepsilon_0 < \frac{\mu\gamma}{2K}. \tag{1}$$

Equivalently, $\gamma/K > 2\varepsilon_0/\mu$: the minimum displacement needed for forgetting exceeds the distance at which the quadratic penalty outweighs the possible first-order improvement.

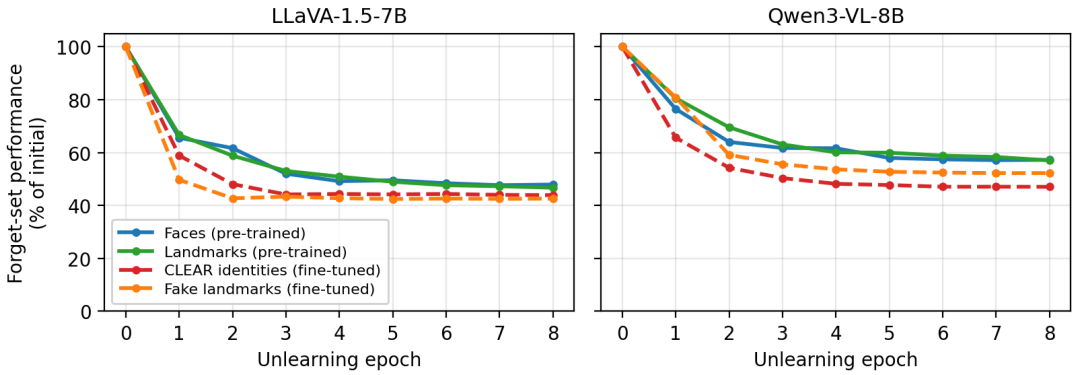


Figure 7: Forget-set image-textual performance (relative to the model before unlearning) over unlearning epochs, averaged over all 8 learning-based unlearning methods (i.e., excluding the pruning-based MANU), on LLaVA-1.5-7B (left) and Qwen3-VL-8B-Instruct (right). Fine-tuned knowledge (dashed) collapses more quickly than pre-trained knowledge (solid).

B.3 Proof

The two cases differ in whether θ_0 already satisfies the forgetting criterion. In part I, returning to θ_0 gives a feasible solution with zero retention cost relative to that same model. In part II, its target score exceeds the threshold by $\gamma > 0$. Successful forgetting then requires moving away from θ_0, and the assumptions translate that required displacement into a positive retention cost.

Fine-tuned knowledge (part I). Let $\theta_{\text{ft}} = \theta_0 + \Delta_{\text{ft}}$, where

$$s_{\text{target}}(\theta_0) \le \tau < s_{\text{target}}(\theta_{\text{ft}}).$$

Reversing the fine-tuning update gives $\theta_u = \theta_{\text{ft}} - \Delta_{\text{ft}} = \theta_0$. This model satisfies the forgetting criterion and has zero retention cost:

$$s_{\text{heldout}}(\theta_0) - s_{\text{heldout}}(\theta_u) = 0.$$

Thus, a zero-cost solution exists without requiring Assumptions A1–A3.

Pre-trained knowledge (part II). Let $\theta_u = \theta_0 + \delta \in \Theta$ satisfy $s_{\text{target}}(\theta_u) \le \tau$. We prove the positive retention-cost bound in three steps.

Step 1: Required displacement. The forgetting criterion requires a score reduction of at least γ. By Assumption A1,

$$\gamma \le s_{\text{target}}(\theta_0) - s_{\text{target}}(\theta_u) \le K\|\delta\|_2.$$

Hence,

$$\|\delta\|_2 \ge \frac{\gamma}{K}. \tag{2}$$

Step 2: Retention cost. Rearranging Assumption A3 and applying Cauchy–Schwarz with Assumption A2 gives

$$\begin{aligned} &s_{\text{heldout}}(\theta_0) - s_{\text{heldout}}(\theta_u) \\ &\quad \ge -\langle \nabla s_{\text{heldout}}(\theta_0), \delta\rangle + \frac{\mu}{2}\|\delta\|_2^2 \\ &\quad \ge \frac{\mu}{2}\|\delta\|_2^2 - \varepsilon_0\|\delta\|_2. \end{aligned} \tag{3}$$

The first term is the quadratic penalty; the second accounts for the largest possible first-order improvement.

Step 3: Positive lower bound. Factoring the right-hand side of (3) gives

$$s_{\text{heldout}}(\theta_0) - s_{\text{heldout}}(\theta_u) \ge \|\delta\|_2 \left(\frac{\mu}{2}\|\delta\|_2 - \varepsilon_0\right).$$

By (2), the second factor satisfies

$$\frac{\mu}{2}\|\delta\|_2 - \varepsilon_0 \ge \frac{\mu\gamma}{2K} - \varepsilon_0 > 0,$$

where the strict inequality follows from condition (1). We now replace the two occurrences of $\|\delta\|_2$ in the factored expression in turn:

$$\begin{aligned} &s_{\text{heldout}}(\theta_0) - s_{\text{heldout}}(\theta_u) \\ &\quad \ge \|\delta\|_2 \left(\frac{\mu\gamma}{2K} - \varepsilon_0\right) \\ &\quad \ge \frac{\gamma}{K}\left(\frac{\mu\gamma}{2K} - \varepsilon_0\right) \\ &\quad = \frac{\mu\gamma^2}{2K^2} - \frac{\varepsilon_0\gamma}{K} > 0. \end{aligned}$$

The first inequality replaces the norm inside the parentheses by its lower bound γ/K, while keeping the positive outer factor $\|\delta\|_2$ unchanged. The

second replaces the outer factor by γ/K; the inequality direction is preserved because the remaining factor $\mu\gamma/(2K) - \varepsilon_0$ is strictly positive. This proves part II for every successful unlearning solution within Θ under the stated conditions.

Practical Interpretation of the Assumptions. Assumptions A1–A3 provide sufficient conditions for a local forgetting–retention trade-off. A1 bounds the sensitivity of the target knowledge score, A2 constrains the gradient of retained utility, and A3 imposes a quadratic retention penalty under parameter changes. We acknowledge that A3 is a strong assumption, and therefore the derived bound should be interpreted as a conditional theoretical result rather than a universal guarantee of retention degradation.

C Hyperparameters and Implementation Details

C.1 Hyperparameter and checkpoint selection

We base our default hyperparameters on MMUnlearner (Huo et al., 2025) and tune each method using only training results. We favor settings that keep retain-set performance high and forget-set performance low, using

$$\text{Selection score} = \text{mean retain-train score} - \text{mean forget-train score}.$$

Higher is better. For each subset, the means give equal weight to classification accuracy, fill-in-the-blank accuracy, and ROUGE-L, and to image-text and text-only inputs. Accuracies are expressed as percentages and ROUGE-L is multiplied by 100, putting all three metrics on a 0–100 scale.

Hyperparameters are selected on training splits only to prevent the obvious test-set leakage. Due to the low number (five) of training images per concept, we do not further split the training set into training and validations sets. [3] For each candidate configuration, we score every checkpoint (epoch) by the difference between its average retain-train performance and its average forget-train performance, where both averages span the six task–modality metrics (three tasks $\times$ VQA/QA). For each of the four real-world subsets (human identities, car brands, fruits, and landmarks), we take the best score across epochs, and the configuration with the highest average of these four scores is selected. Under the selected configuration, the reported checkpoint is then chosen per subset as the epoch with the highest score. The synthetic subsets (CLEAR and fake landmarks) reuse the selected configuration and choose their own epochs, but do not influence the configuration choice.

For both LLaVA-1.5-7B and Qwen3-VL-8B-Instruct experiments, we use a batch size of 2, random seed 42, and 8 training epochs. LoRA uses rank 16, scaling parameter 16, and dropout 0.05.

[3] For benchmarks that have a larger number of training images, it would be beneficial to create such a validation set.

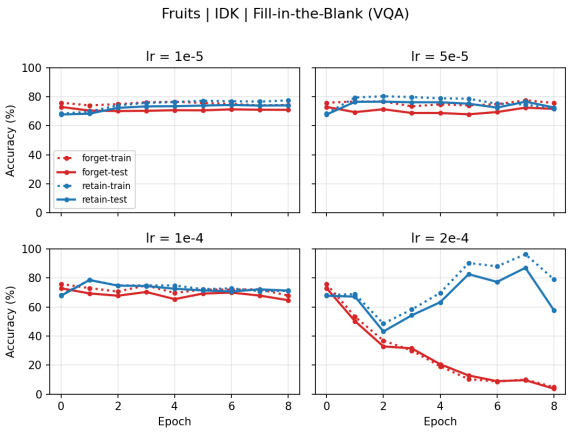


Figure 8: Hyperparameter sensitivity within a single method. IDK on the Fruits subset (Qwen3-VL-8B, fill-in-the-blank, VQA) across its learning-rate grid. Three of the four learning rates leave forget-set accuracy essentially unchanged after eight epochs.

D More Examples on Hyperparameter Sensitivity

Figure 8 shows a further example of the sensitivity, this time within a single method. We run IDK on the Fruits subset of Qwen3-VL-8B with four learning rates spanning its search grid. At $\text{lr} \in \{10^{-5}, 5 \times 10^{-5}, 10^{-4}\}$, forget-set accuracy remains essentially at the pre-trained level after eight epochs, i.e., no unlearning occurs. Only at $\text{lr} = 2 \times 10^{-4}$ does the method succeed.

E Dataset Examples

E.1 Representative Examples

Figures 9–14 show representative examples from all six subsets. Each figure retains the forget/retain $\times$ train/test layout, and each panel separates image-text questions (VQA) from text-only questions (QA). Only the VQA input includes the displayed image; QA names the concept in its text and receives no image. VQA and QA examples are presented to the model as separate inputs.

The examples probe associated knowledge, including biographical details, literary works, landmark completion dates, brand history, and fruit attributes. Each panel presents one VQA and one QA example; across the four panels of each subset, we illustrate single-choice classification (CLS), fill-in-the-blank (FITB), and generation (GEN). Both modalities support all three task types; the figures show selected combinations. Questions, options, and reference answers are taken from the corresponding dataset rows.

Within each forget or retain row, train and test use different images of the same concept and probe the same knowledge with rephrased questions. This applies to both VQA and QA. For CLS, test distractors and option order can also change, so we show both the correct option letter and its answer text.

Forget – Train

VQA CLS

Question: What is this person's nationality?

A. Colombia **B.** Greece
C. United States **D.** Germany

Correct answer: C. United States

QA GEN

Question: When and where was Brad Pitt born?

Reference answer: Brad Pitt was born in Shawnee, Oklahoma, United States on December 18, 1963.

Forget – Test

VQA CLS

Question: What is the nationality of the person shown?

A. Brazil **B.** United States
C. Ireland **D.** Serbia

Correct answer: B. United States

QA GEN

Question: What are the birth date and birthplace of Brad Pitt?

Reference answer: Brad Pitt was born in Shawnee, Oklahoma, United States on December 18, 1963.

Retain – Train

VQA GEN

Question: What is this person best known for?

Reference answer: This person is best known for her self-deprecating stand-up comedy.

QA FITB

Prompt: Amy Schumer was born in New York City, New York, United States on June 1, ___.

Answer: 1981

Retain – Test

VQA GEN

Question: Why is this person famous?

Reference answer: This person is best known for her self-deprecating stand-up comedy.

QA FITB

Prompt: The birth year of Amy Schumer is ___.

Answer: 1981

Figure 9: Human identities. Paired train/test examples illustrate rephrasing for both VQA and text-only QA. CLS: single-choice classification; FITB: fill-in-the-blank; GEN: generation. Answers are dataset references.

Forget – Train

VQA FITB

Prompt: This person is best known for her Western works, including '___' and 'Shea's Grudge', and for winning the Red Saddle Award.

Answer: Bingham's Standoff

QA GEN

Question: When and where was Adrianne Lebeau born?

Reference answer: Adrianne Lebeau was born in Brussels, Belgium on July 16, 1965.

Forget – Test

VQA FITB

Prompt: This person is widely known for her Western works, including '___' and 'Shea's Grudge', and for winning the Red Saddle Award.

Answer: Bingham's Standoff

QA GEN

Question: What are the birth date and birthplace of Adrianne Lebeau?

Reference answer: Adrianne Lebeau was born in Brussels, Belgium on July 16, 1965.

Retain – Train

VQA CLS

Question: What is this person's occupation?

A. fantasy author
B. humour author
C. true crime author
D. werewolf fiction author

Correct answer: C. true crime author

QA FITB

Prompt: Adetoun Davis was born in ___, Nigeria on January 16, 1941.

Answer: Lagos

Retain – Test

VQA CLS

Question: State this person's profession.

A. true crime author
B. Western author
C. humour author
D. Portuguese literature author

Correct answer: A. true crime author

QA FITB

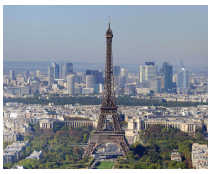
Prompt: The birthplace of Adetoun Davis is ___, Nigeria.

Answer: Lagos

Figure 10: CLEAR. Paired train/test examples illustrate rephrasing for both VQA and text-only QA. CLS: single-choice classification; FITB: fill-in-the-blank; GEN: generation. Answers are dataset references.

Forget – Train

VQA CLS



Question: In what year was this landmark completed?
A. 1889 **B.** 1976
C. 1941 **D.** 966
Correct answer: A. 1889

QA FITB

Prompt: The Eiffel Tower is located in the city of ___.
Answer: Paris

Forget – Test

VQA CLS

Question: In which year was construction of this landmark completed?
A. 1905 **B.** 1645
C. 1889 **D.** 1932
Correct answer: C. 1889

QA FITB

Prompt: The Eiffel Tower is in the city of ___.
Answer: Paris

Retain – Train

VQA GEN

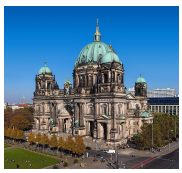
Question: In what year was this landmark completed?
Reference answer: This landmark was completed in 1905.

QA CLS

Question: What country is Berlin Cathedral located in?
A. Italy **B.** Canada
C. France **D.** Germany
Correct answer: D. Germany

Retain – Test

VQA GEN



Question: In what year did this landmark reach completion?
Reference answer: This landmark was completed in 1905.

QA CLS

Question: In which country can you find Berlin Cathedral?
A. Belgium **B.** United States
C. Spain **D.** Germany
Correct answer: D. Germany

Figure 11: Landmarks. Paired train/test examples illustrate rephrasing for both VQA and text-only QA. CLS: single-choice classification; FITB: fill-in-the-blank; GEN: generation. Answers are dataset references.

Forget – Train

VQA FITB

Prompt: This landmark was completed in ___.
Answer: 2025

QA CLS

Question: What city is the Opal Tower located in?
A. Quito **B.** Sydney
C. Vienna **D.** Accra
Correct answer: B. Sydney

Forget – Test

VQA FITB

Prompt: This landmark reached completion in the year ___.
Answer: 2025

QA CLS

Question: Which city is the Opal Tower located in?
A. Sydney **B.** Mendoza
C. Kyoto **D.** Tashkent
Correct answer: A. Sydney

Retain – Train

VQA CLS

Question: What country is this landmark located in?
A. Greece **B.** Italy
C. Austria **D.** Denmark
Correct answer: C. Austria

QA GEN

Question: In what year or century was the Scarlet Whispers Abbey completed?
Reference answer: The Scarlet Whispers Abbey was completed in the 11th century.

Retain – Test

VQA CLS

Question: Which country does this landmark stand in?
A. Austria **B.** Czech Republic
C. Iceland **D.** United Kingdom
Correct answer: A. Austria

QA GEN

Question: What year or century saw the completion of the Scarlet Whispers Abbey?
Reference answer: The Scarlet Whispers Abbey was completed in the 11th century.

Figure 12: Synthetic landmarks. Paired train/test examples illustrate rephrasing for both VQA and text-only QA. CLS: single-choice classification; FITB: fill-in-the-blank; GEN: generation. Answers are dataset references.

Forget – Train

VQA CLS

Question: What year was this brand founded?
A. 1916 **B.** 1904
C. 1909 **D.** 1970
Correct answer: A. 1916

QA GEN

Question: What country is BMW headquartered in?
Reference answer: BMW is headquartered in Germany.

Forget – Test

VQA CLS

Question: When was this brand founded?
A. 1937 **B.** 1902
C. 1916 **D.** 1926
Correct answer: C. 1916

QA GEN

Question: Which country is BMW based in?
Reference answer: BMW is headquartered in Germany.

Retain – Train

VQA GEN

Question: What is this brand's parent company?
Reference answer: The parent company of this car brand is Stellantis.

QA FITB

Prompt: Opel was founded in the year ___.
Answer: 1862

Retain – Test

VQA GEN

Question: Name the company that owns this car brand.
Reference answer: The parent company of this car brand is Stellantis.

QA FITB

Prompt: The year in which Opel was founded is ___.
Answer: 1862

Figure 13: Car brands. Paired train/test examples illustrate rephrasing for both VQA and text-only QA. CLS: single-choice classification; FITB: fill-in-the-blank; GEN: generation. Answers are dataset references.

Forget – Train

VQA FITB

Prompt: This fruit originated in ___.
Answer: Central Asia

QA CLS

Question: What climate zone is the apple grown in?
A. subtropical **B.** arid
C. tropical **D.** temperate
Correct answer: D. temperate

Forget – Test

VQA FITB

Prompt: This fruit first came from ___.
Answer: Central Asia

QA CLS

Question: In which climate zone is the apple cultivated?
A. tropical **B.** temperate
C. subtropical **D.** mediterranean
Correct answer: B. temperate

Retain – Train

VQA GEN

Question: When is this fruit in season?
Reference answer: This fruit is in season during autumn (October to November).

QA FITB

Prompt: The persimmon grows on a ___.
Answer: tree

Retain – Test

VQA GEN

Question: At what time of year is this fruit in season?
Reference answer: This fruit is in season during autumn (October to November).

QA FITB

Prompt: The persimmon is borne on a ___.
Answer: tree

Figure 14: Fruits. Paired train/test examples illustrate rephrasing for both VQA and text-only QA. CLS: single-choice classification; FITB: fill-in-the-blank; GEN: generation. Answers are dataset references.

Method	Learning rate search space	Method-specific settings
GA	$\{10^{-6}, \mathbf{2 \times 10^{-6}}, 5 \times 10^{-6}, 10^{-5}\}$	—
GADiff	$\{10^{-6}, 2 \times 10^{-6}, 5 \times 10^{-6}, \mathbf{10^{-5}}\}$	—
KL Min	$\{2 \times 10^{-6}, 5 \times 10^{-6}, 10^{-5}, \mathbf{2 \times 10^{-5}}\}$	—
IDK	$\{10^{-5}, 5 \times 10^{-5}, 10^{-4}, \mathbf{2 \times 10^{-4}}\}$	—
LLMU	$\{10^{-5}, \mathbf{2 \times 10^{-5}}, 5 \times 10^{-5}, 10^{-4}\}$	—
NPO	$\{10^{-6}, 2 \times 10^{-6}, 5 \times 10^{-6}, \mathbf{10^{-5}}\}$	$\beta \in \{\mathbf{0.025}, 0.1, 0.4\}$
SCRUB	$\{10^{-5}, 2 \times 10^{-5}, 5 \times 10^{-5}, \mathbf{10^{-4}}\}$	Forget-KL weight $\beta \in \{0.1, \mathbf{1.0}, 2.0\}$
RMU	$\{5 \times 10^{-6}, 10^{-5}, 2 \times 10^{-5}, \mathbf{5 \times 10^{-5}}\}$	Steering coefficient $c \in \{20, \mathbf{100}, 300\}$, layer $= 7$, forget-loss weight $= 1.0$
MANU	—	Prune ratio $\in \{0.5, 1, 2, \mathbf{5}\}\%$

Table 4: Hyperparameter search space and selected values (**bold**) for Qwen3-VL-8B-Instruct. Learning rates are swept with method-specific settings fixed at their selected values, and method-specific settings are swept at a learning rate of 10^{-5}.

Method	Learning rate search space	Method-specific settings
GA	$\{\mathbf{10^{-6}}, 5 \times 10^{-6}, 10^{-5}, 5 \times 10^{-5}\}$	—
GADiff	$\{10^{-6}, 5 \times 10^{-6}, 10^{-5}, \mathbf{5 \times 10^{-5}}\}$	—
KL Min	$\{10^{-6}, 5 \times 10^{-6}, 10^{-5}, \mathbf{5 \times 10^{-5}}\}$	—
IDK	$\{10^{-5}, 5 \times 10^{-5}, 10^{-4}, \mathbf{2 \times 10^{-4}}\}$	—
LLMU	$\{10^{-5}, 2 \times 10^{-5}, \mathbf{5 \times 10^{-5}}, 10^{-4}\}$	—
NPO	$\{10^{-6}, 5 \times 10^{-6}, \mathbf{10^{-5}}, 5 \times 10^{-5}\}$	$\beta \in \{0.025, 0.1, \mathbf{0.4}\}$
SCRUB	$\{10^{-5}, 2 \times 10^{-5}, 5 \times 10^{-5}, \mathbf{10^{-4}}\}$	Forget-KL weight $\beta \in \{0.1, \mathbf{1.0}, 2.0\}$
RMU	$\{5 \times 10^{-6}, 10^{-5}, 2 \times 10^{-5}, \mathbf{5 \times 10^{-5}}\}$	Steering coefficient $c \in \{20, \mathbf{100}, 300\}$, layer $= 7$, forget-loss weight $= 1.0$
MANU	—	Prune ratio $\in \{0.5, 1, 2, \mathbf{5}\}\%$

Table 5: Hyperparameter search space and selected values (**bold**) for LLaVA-1.5-7B. Learning rates are swept with method-specific settings fixed at their selected values, and method-specific settings are swept at a learning rate of 10^{-5}.